ISSUE 01 · POLAR / ICE CITY · A CITY BUILT IN 3D

# AURORA HAVEN

*A polar city that thrives*



## The city's own sun

A compact fusion station keeps 45,000 people warm through the polar night

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## A warm heart inside the ice

Four geothermal spa levels hidden in a 35-metre snow peak

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## Bread and water from snow and air

Inside the plant that feeds a city with no farmland

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## −45 °C. 120 km/h. No sun.

The polar challenge, and the five ideas that answer it

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# *“Our goal was not only to survive the cold but to build a city where people really want to live.”*



*The Aurora Lagoon, a 38 °C open-air mineral pool at the foot of Mount Aurora, under the northern lights.*

**WELCOME TO THIS SPECIAL FEATURE** on a polar city named Aurora Haven – a bold architectural marvel where human ingenuity meets the call of the wild. In an era shaped by global climate shifts, this project is more than just an urban concept; it represents a strategic vision for survival and sustainable development in the earth’s most extreme polar environments.

From its sleek structural design to intelligent renewable energy integration, Aurora Haven redefines how humanity can adapt and thrive under pressure. We invite you to turn the page and step into a world where extreme natural challenges are transformed into a vibrant, resilient, and forward-thinking community.

***The Polar City team***

Class 11A4 · Lê Quý Đôn High School



ON THE COVER

Aurora 81 – Polar Landmark Tower, named in honour of Landmark 81 in Ho Chi Minh City, beside the street that leads to the glowing terraces of the Mount Aurora spa, under the northern lights. Its nine bundled glass tubes step back like bamboo stalks and cut blizzard wind load by about 25%.

THE TEAM BEHIND AURORA HAVEN

# Meet the Founders

*The people who planned, built and wrote Aurora Haven, and what each of us did.*



### Nguyễn Đông Bảo

Magazine design · 3D city design



### Nguyễn Bá Hiếu Thiện

Team leader · Research · 3D city design



### Phan Minh Khôi

Research · Fact-checking



### Dương Hàng Nguyên

Video filming & editing · 3D city design



### Phạm Phước Khoa

Video editing · Magazine design



### Nguyễn Minh Triết

Magazine design · Report

SCAN TO WATCH

## Our real-world interview

Video on YouTube
youtu.be/QwcP_2Q3Esc



CITY OVERVIEW

# Welcome to Aurora Haven

*Designed not just to survive the polar night, but to make life here warm, healthy and sustainable.*

**AURORA HAVEN IS** a compact polar city of around 45,000 people, built on a frozen tundra plateau beneath the snow peak of Mount Aurora. Winters fall to −45 °C, the sun disappears for two months of polar night, and almost nothing can grow.

The whole city is shaped around one idea: heat, energy, water and food should never be wasted. Every building shares a single heating network, the city runs on zero-carbon energy, and nearly all of its water and waste is recycled.

Homes, schools, healthcare, shops and parks sit within a short walk of each other, connected by a downtown metro, electric trams and heated streets.



*Aurora Haven at dusk, seen from the south-east: the fusion station in front, downtown and Aurora 81 in the middle, the wind farm on the windy northern edge.*

What makes Aurora Haven special is its three signature districts. A fusion power station acts as the city's own sun, giving steady power and heat even through the polar night. A Snow-to-Food & Water Plant turns snow and air into clean drinking water and food, so the city can feed itself without farmland.

And inside Mount Aurora, a geothermal spa resort offers warmth, rest and a view of the northern lights. Together they form one closed loop that keeps the city powered, fed and happy in one of the harshest places on Earth.

**45,000**
residents

**−45 °C**
winter lows

**35 m**
Mount Aurora, the snow peak

**10 min**
walk to daily needs



THE PLACE

# The Polar Challenge

*One of the hardest places on Earth to build a city, so every building had to answer one question.*

Aurora Haven sits in one of the hardest environments on Earth. In winter the temperature falls to minus 45 degrees. For about two months the sun never rises, as in Tromsø, Norway, which sees no sunrise from late November to mid-January [19]. Blizzards can reach 120 kilometres per hour.

The ground is permafrost, so a warm building can thaw it and crack its own foundations. Across the Arctic, about 70% of today's buildings, roads and pipelines on permafrost stand where the ground could thaw by 2050 [2]. There is almost no farmland, and in deep winter supplies can only arrive by air.

So we asked one question about every building in the city:

***"How does it help people stay warm, fed, safe and happy here?"***

**−45 °C**

winter temperatures

**2 months**

of polar night, and two of midnight sun

**120 km/h**

blizzards

**≈4×**

faster warming in the Arctic than the world since 1979 [1]

*A blizzard sweeps past the snowbreak pines towards the Ice-Proof Wind Farm on the windy northern edge.*



*Downtown in the low sun, with Aurora 81, the food plant silos and Mount Aurora behind.*

THE VISION

# Our Vision & Mission

***Our mission: a zero-carbon city where 45,000 people stay warm, fed, safe and happy through the polar night, without harming the tundra.***

## WHO IS AFFECTED?

### 45,000 residents

Families, children, older people and shift workers face −45 °C, two months without sun and 120 km/h blizzards every winter.

### 4 million Arctic people

About 4 million people live in the Arctic today [3]. By 2050, nearly 4 million people may live where thawing ground can crack homes, roads and pipes [2].

### Supply chains

In deep winter, food, medicine and spare parts can only arrive by air, so one long storm can cut a city off.

### The tundra and its wildlife

Arctic foxes, hares, ptarmigans and reindeer depend on fragile, frozen land that a careless city would pollute, light up or melt.

## FIVE DESIGN PRINCIPLES

01 **Heat is life:** one heating loop, fed by fusion and geothermal heat, warms every building; insulated piles keep the permafrost frozen.

02 **Zero carbon:** no fossil fuel burns anywhere; power comes from fusion, wind, sun and geothermal heat.

03 **Closed loops:** about 95% of water is recycled and rubbish is reused instead of thrown away.

04 **Compact and car-light:** daily needs within a 10-minute walk; the few cars are electric.

05 **Resilient and healthy:** shelters for 30 days, and light, parks, culture and a spa for the polar night.




DOWNTOWN

# Nine Blocks, One Short Walk

*Shops, schools, healthcare and homes within a short walk of each other, around the Aurora Roundabout.*

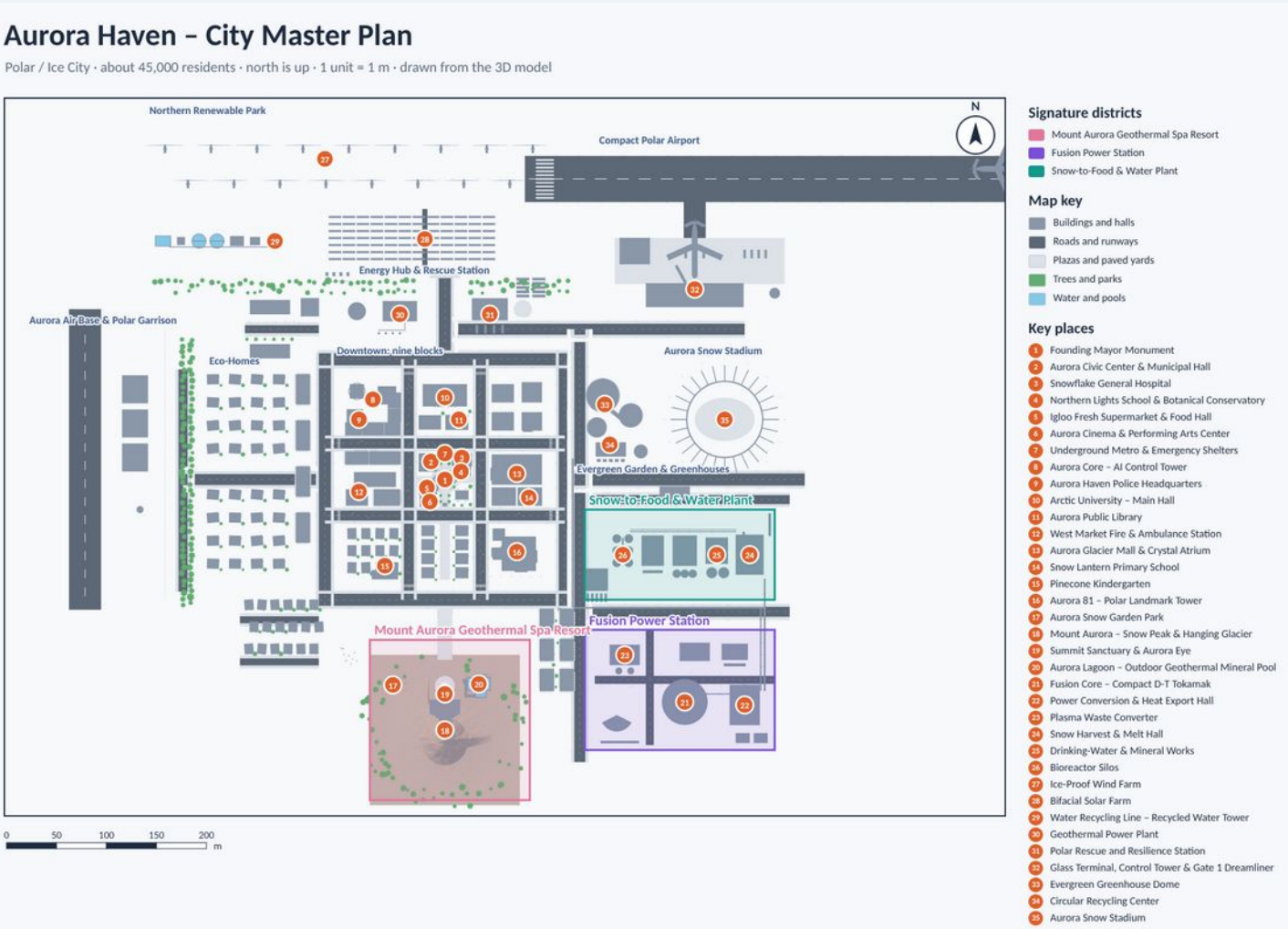
Downtown is laid out in nine blocks around the Aurora Roundabout and the Founding Mayor Monument. The metro interchange lies under Liberty Mall, the tree-lined promenade through the four central blocks, which hosts festivals, markets and parades.



*Liberty Mall at dusk, looking north to the Founding Mayor Monument. Its granite plaza is warmed from below by the heating loop and never ices, and twin lanterns line the way.*

THE MAP

# The Whole City Plan




*Aurora Haven – City Master Plan. North is up; one unit is one metre. Drawn from the 3D model; the numbers are the 35 places listed on page 36.*

### DOWNTOWN

Nine blocks around the Aurora Roundabout and the Founding Mayor Monument, with the hospital, schools, shops and the metro interchange.

### WEST

The air base and garrison, behind a belt of trees that screens the eco-homes.

### SOUTH

Framed by Liberty Mall, Mount Aurora rises with its geothermal spa resort inside.

### NORTH

The windy north holds the Northern Renewable Park, the energy hub, the Polar Rescue and Resilience Station and the airport.

### EAST

Greenhouses, the snow stadium, the fusion station and the Snow-to-Food plant, linked by pipe bridges and heat mains.

DOWNTOWN

# The Nine Blocks



*Downtown from the south-east. The map below follows the city plan, with north at the top.*

### Polaris Quarter

Aurora Core – AI Control Tower · Aurora Haven Police Headquarters · Polaris Green Office Tower

### Knowledge Quarter

Arctic University – Main Hall · Aurora Public Library

### Financial District

Arctic Stock Exchange · Aurora Green Finance Hall · Snowline Courthouse · Aurora Business Hotel

### West Market Street

West Market Fire & Ambulance Station · Aurora Night Market Hall · Polar Pharmacy

### Civic Core & Liberty Mall

Founding Mayor Monument · Civic Center · Snowflake General Hospital · Northern Lights School · Igloo Fresh · Aurora Cinema · Metro

### East Gallery Avenue

Aurora Glacier Mall & Crystal Atrium · Snow Lantern Primary School · Northern Lights Music Hall

### Snowpine Residences

Family homes · Pinecone Kindergarten · Snowpine Senior Care Home

### Mount Aurora Approach

Mount Aurora Guesthouse · Southside Police & Mountain Rescue Post · spa shuttle street

### Aurora 81 District

Aurora 81 – Polar Landmark Tower · Aurora 81 Residences (Twin Towers)

1

SIGNATURE DISTRICT 1 · FUSION POWER STATION

# The City's Own Sun

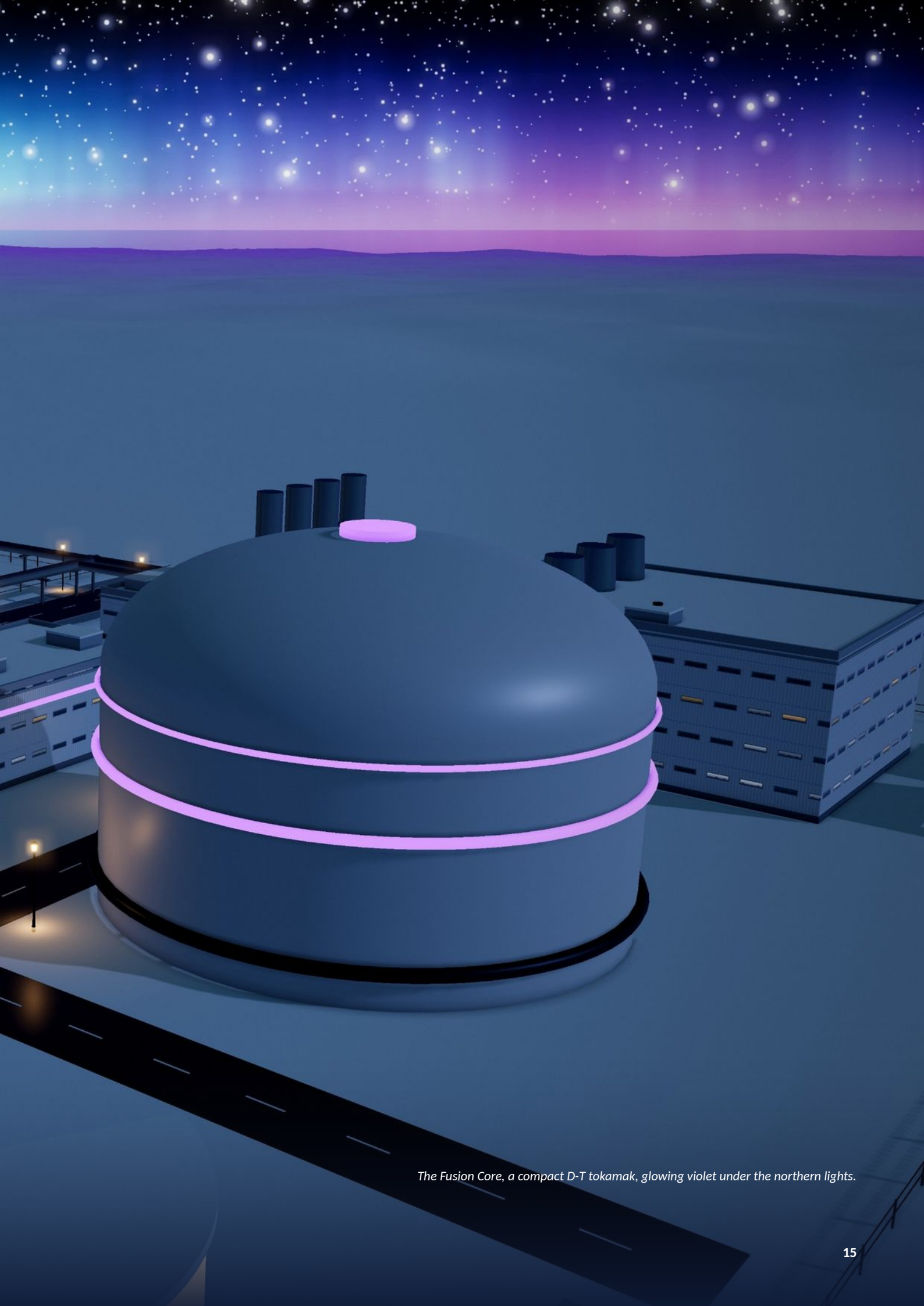
*A compact tokamak gives Aurora Haven about 300 MW of electricity and 450 MW of district heat, steady through the polar night, with no $CO_2$ and no meltdown.*



Our first signature district is the city's own sun. Stage 1 takes deuterium from snowmelt water, so the fuel literally falls from the sky as snow. Stage 2 breeds tritium in a lithium blanket, so none is imported.

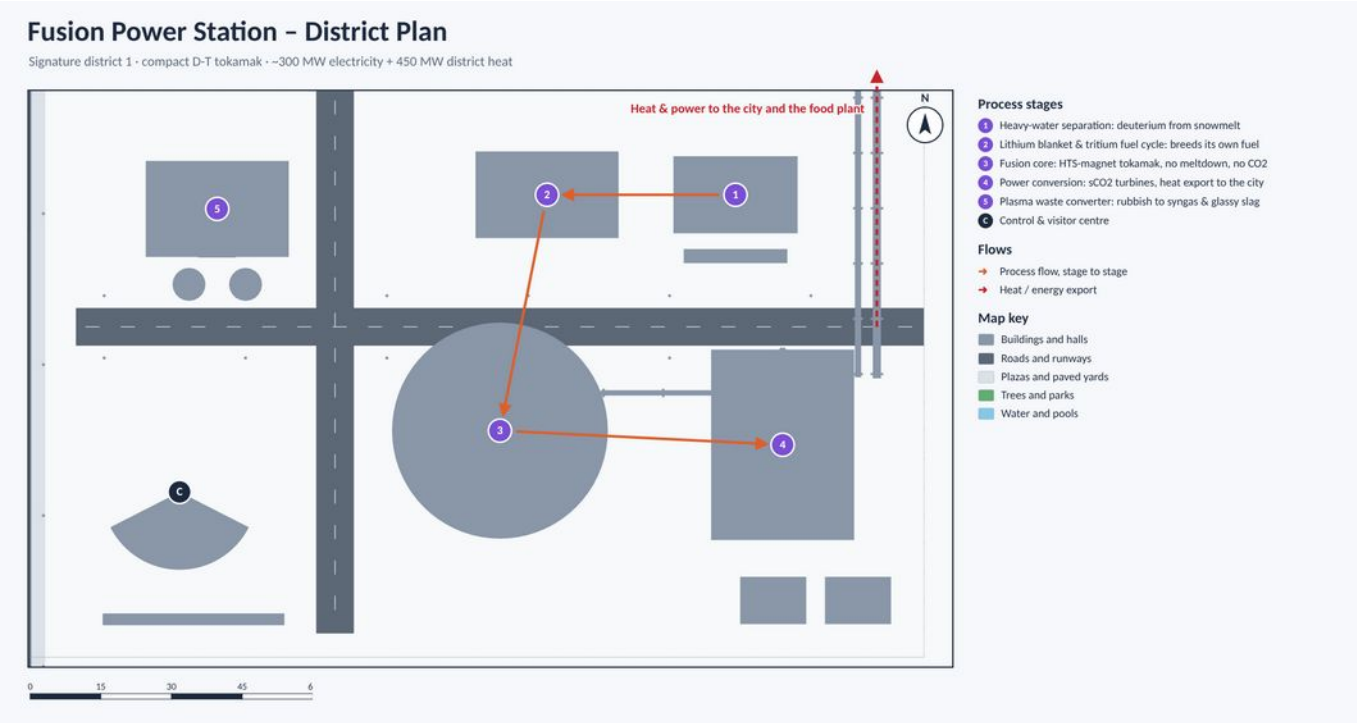
Stage 3 is the tokamak, where superconducting magnets hold a 150-million-degree plasma that makes about 800 megawatts of heat. Stage 4 turns that heat into about 300 megawatts of electricity and sends 450 megawatts into the district-heating loop.

The reactor runs on about 170 grams of fuel a day. If anything fails, the plasma simply goes out in a split second, so it cannot melt down or run away.



*The Fusion Core, a compact D-T tokamak, glowing violet under the northern lights.*

# Five Stages, One Sun




*Fusion Power Station – District Plan. Signature district 1: a compact D-T tokamak.*

**1** **Heavy-water separation**

Takes deuterium from the deuterium-rich water left by the food plant's electrolysers and purifies it to 99.8% heavy water.

**2** **Lithium blanket & tritium fuel cycle**

Breeds slightly more tritium than the reactor burns, so none is imported. Recovered helium cools the hospital's MRI scanners.

**3** **Fusion core: compact D-T tokamak**

Superconducting magnets hold a 150-million-degree plasma. No $CO_2$, no long-lived waste, no meltdown.

**4** **Power conversion & heat export**

Supercritical-$CO_2$ turbines make about 300 MW of electricity and send about 450 MW of heat to the city.

**5** **Plasma waste converter**

Torches hotter than 5,000 °C turn about 20 t of rubbish a day into clean gas, metals and paving blocks.

**C** **Control & visitor centre**

Operators and an AI digital twin can stop the plasma in milliseconds; a public gallery faces the reactor.

**800 MW**
of heat from the plasma

**300 MW**
of electricity

**450 MW**
of district heat for the city

**170 g**
of fuel a day, about 70 g of it deuterium

**1 litre**
of liquid helium a day for MRI scanners

2

SIGNATURE DISTRICT 2 · SNOW-TO-FOOD & WATER PLANT

# Bread and Water from Snow and Air

*Six stages turn tundra snow and air into all of the city's drinking water and about half of its protein, with no farmland at all.*

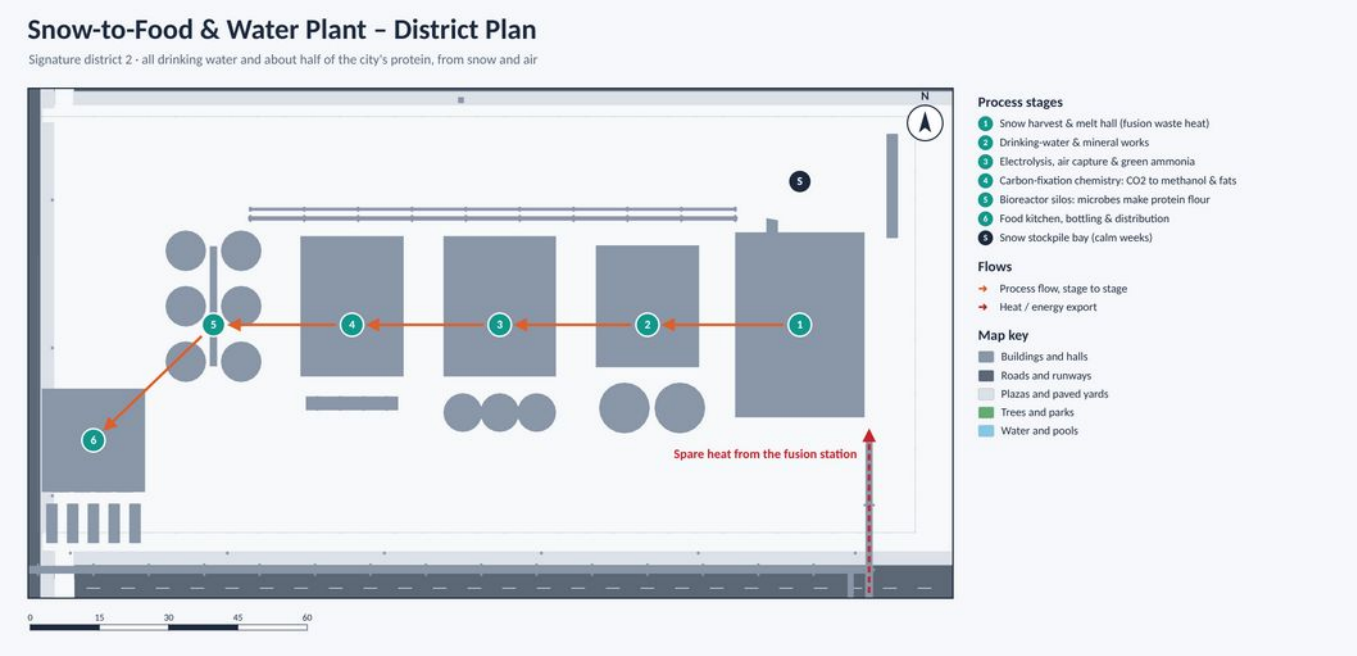


*The six Bioreactor Silos in the low sun.*

What do you eat where nothing grows? Stage 1 melts about 3,000 tonnes of clean tundra snow a day with spare heat from the fusion station, so the snow does the job of a cooling tower. Stage 2 turns the meltwater into all the city's drinking water.

Stage 3 splits water into hydrogen and oxygen and captures $CO_2$ from the air; Stage 4 turns them into starch and fats. In Stage 5, microbes in six silos make about 2.5 tonnes of protein flour a day. Stage 6 cooks bread, noodles and drinks for the city.

# Six Stages from Snow to Supper




*Snow-to-Food & Water Plant – District Plan. All drinking water and about half of the city's protein, from snow and air.*

**1** **Snow harvest & melt hall**

Electric harvesters gather wind-packed snow upwind of the city; about 14 MW of spare fusion heat melts it.

**2** **Drinking-water & mineral works**

Membranes and UV light remove every germ, calcite adds calcium and magnesium, and sensors test the water every five minutes.

**3** **Electrolysis, air capture & green ammonia**

About 5 t of hydrogen a day; the oxygen goes to the hospital and the waste converter.

**4** **Carbon-fixation chemistry**

A copper catalyst makes methanol, and enzymes build it into starch about eight times faster than a maize field.

**5** **Bioreactor silos**

Hydrogen-eating bacteria make protein flour; yeasts and algae make oils and vitamins, 24 hours a day.

**6** **Food kitchen, bottling & distribution**

Bread, noodles, protein mince and dairy-style drinks, delivered overnight in heated electric vans.

**3,000 t**

of snow melted a day

**2,500 m³**

of drinking water a day, all of the city's tap water

**20 t**

of $CO_2$ captured from the air a day

**2.5 t**

of protein flour a day, as much as 600 hectares of soybeans

**30 days**

of meals and water for everyone in the Food Reserve Bunker

3

SIGNATURE DISTRICT 3 · MOUNT AURORA GEOTHERMAL SPA RESORT

# A Warm Heart Inside the Ice

*Inside a 35-metre snow peak shaped like Nepal's Ama Dablam, four geothermal spa levels stay at 28 °C all winter.*



*The four spa levels step out of the north face; the forecourt is warmed by the spa's used hot water, so guests walk ice-free from the shuttle to the door.*

Guests enter through the North Portal, where a thermal revolving door and an air curtain keep the −45 °C air out. The peak's deep snow works as a blanket that holds in the geothermal heat, and sensors release the snow load safely before an avalanche can build.

Outside is the Aurora Lagoon, a 38-degree mineral pool for bathing under the northern lights. The spa fights the winter blues, brings tourists and income, and in an emergency can shelter up to 1,000 people.

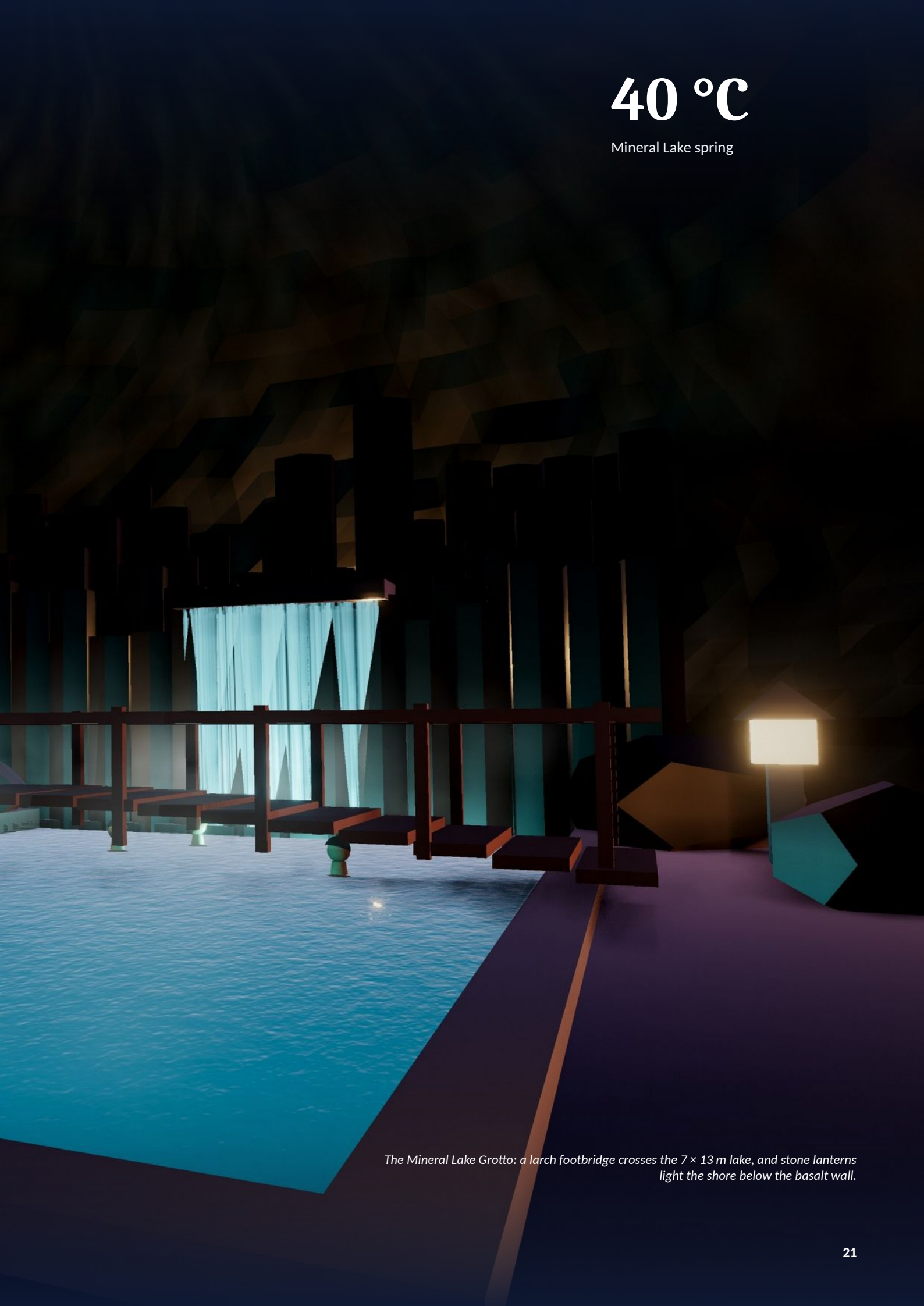
MOUNT AURORA · LEVEL 1

# Inside the Mountain

*The ground level and arrival floor of the spa: a welcome hall, two changing suites, a 7 × 13 m geothermal Mineral Lake grotto and an onsen garden with hot, warm and glacier-cold pools.*

A 40 °C silica- and bicarbonate-rich spring rises from the geothermal field into the lake, spills over a basalt wall and flows on to the outdoor Aurora Lagoon.





**40 °C**

Mineral Lake spring

*The Mineral Lake Grotto: a larch footbridge crosses the 7 × 13 m lake, and stone lanterns light the shore below the basalt wall.*

# Four Floors of Warmth




*Mount Aurora Geothermal Spa Resort – Level Plans. The glass lift links all four levels.*

LEVEL 1 · 0 M

### Welcome Hall, Mineral Lake Grotto & Onsen Garden

A 40 °C silica-rich spring rises into a 7 × 13 m Mineral Lake; hot, warm and glacier-cold pools boost circulation.

LEVEL 2 · 4.5 M

### Volcanic Mud Terraces

An 8 × 10 m mud field held at 39 °C. The mud is filtered and UV-sterilised every night and reused for months.

LEVEL 3 · 9 M

### Treatment Suites & Himalayan Salt Sauna

Hot-stone massage, arctic kelp wraps and a 75 °C pink salt sauna heated by the geothermal loop.

LEVEL 4 · 13.5 M

### Summit Sanctuary & Aurora Eye

A 35 °C float pool, a meditation circle and a heated glass bubble for watching the northern lights.

**28 °C** inside the spa

**40 °C** Mineral Lake spring

**39 °C** volcanic mud

**75 °C** salt sauna

**35 °C** float pool

**38 °C** Aurora Lagoon

*Route through the spa: North Portal → grotto → Welcome Hall → changing suites → Mineral Lake & Onsen Garden → glass lift → mud terraces → treatments & salt sauna → Summit Sanctuary and the Aurora Eye.*

# Three Breakthroughs, Checked

*For each idea: the problem it solves, how it works, what is new compared with today's technology, the science and maths behind it, how realistic it is, and what it changes.*



**1 · Fusion Power Station**



**2 · Snow-to-Food & Water Plant**



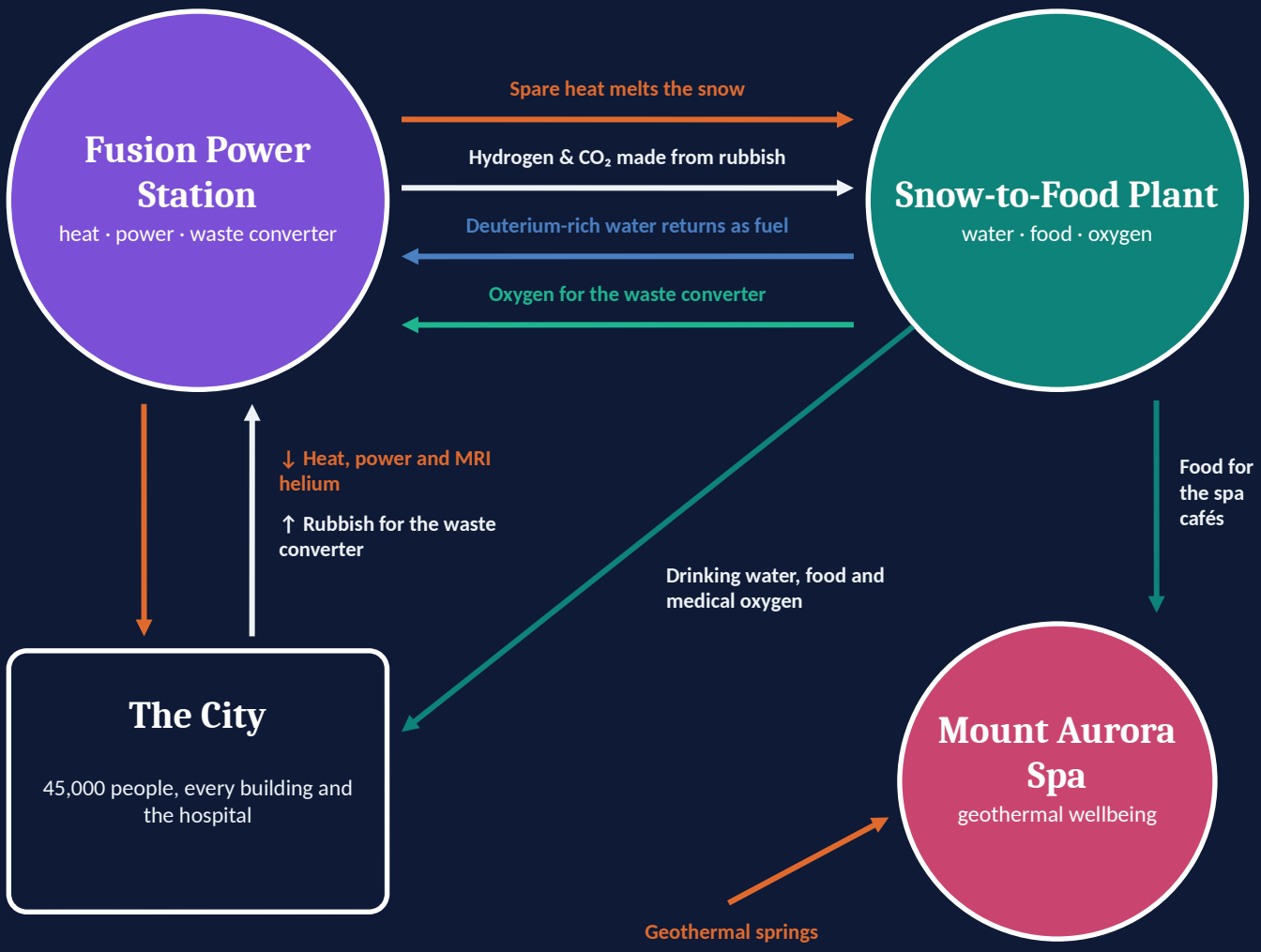
**3 · Mount Aurora Spa**

| | 1 · Fusion Power Station | 2 · Snow-to-Food & Water Plant | 3 · Mount Aurora Spa |
|---|---|---|---|
| THE PROBLEM | Two months without sun and winters of −45 °C. The city needs steady, zero-carbon heat and power every hour of the year. | No farmland, and in deep winter supplies can only arrive by air. How do 45,000 people get water and food? | Months of darkness and cold keep people indoors, and a city needs a warm public place to rest, meet and shelter in a disaster. |
| HOW IT WORKS | A compact D-T tokamak: superconducting magnets hold a 150-million-degree plasma. 800 MW of heat becomes 300 MW of electricity and 450 MW of district heat (page 16). | Spare fusion heat melts 3,000 t of snow a day into drinking water. Electrolysis and air capture make hydrogen, oxygen and $CO_2$, which become starch, fats and microbial protein (page 18). | A 40 °C geothermal spring fills a 7 × 13 m Mineral Lake inside an insulated rock shell that keeps four spa levels at 28 °C; the overflow feeds the 38 °C lagoon outside (page 22). |
| WHAT IS NEW | Fusion machines are still experiments: ITER plans its first D-T plasma in 2039 [4], and the compact SPARC tokamak aims for first plasma around 2027 [5]. Our idea: put one inside a city, so its waste heat warms homes and melts snow for food. | Each step exists alone: Solein protein from hydrogen-eating microbes has been made in Finland since 2024 [6], starch has been made from $CO_2$ in a lab [7], and direct air capture runs in Iceland [8]. Our idea: link them in one plant, run on heat that would be wasted. | Iceland's Blue Lagoon bathes visitors in 37–39 °C water from a geothermal power plant [9], and Switzerland's Therme Vals is built into a hillside [10]. Our idea: a whole spa inside a snow peak that doubles as a warm refuge for 1,000 people. |
| STEAM CHECK | Maths: 800 MW for a day is $6.9 \times 10^{13}$ J. One kilogram of D-T fuel releases about $3.4 \times 10^{14}$ J, so the plasma burns about 0.2 kg a day; heat from the lithium blanket brings this close to our 170 g. Useful energy: (300 + 450) ÷ 800 ≈ 94%. | Physics: melting ice takes 334 kJ/kg, so 3,000 t a day needs about $1.0 \times 10^{12}$ J, or 11.6 MW; warming the snow from −20 °C adds about 1.5 MW. Total ≈ 13 MW, inside the 14 MW of spare fusion heat. 2,500 $m^3$ a day is 55 L per person. | Engineering: the shell holds a 73 °C difference, −45 °C outside and 28 °C inside. Science: the water is used twice, 40 °C in the lake, then 38 °C in the lagoon, before it returns to the geothermal loop. Arts: the peak's shape comes from Nepal's Ama Dablam. |
| FEASIBILITY & LIMITS | A future concept: no fusion plant sells power today, and breeding tritium at scale is still unproven [11]. Costs are huge: ITER alone is estimated at €18–22 billion to first plasma [12]. Plan B: the geothermal plant and a 12-hour heat tank. | Possible in parts today, hard at city scale: Solar Foods' first factory makes up to 160 t a year [6], $CO_2$-to-starch is still lab-scale, and direct air capture is energy-hungry and costly [13]. New foods need safety approval and people's trust. | Needs a hot geothermal source under the site, to be confirmed by drilling. Building inside a frozen mountain is costly, and no heat may reach the permafrost: thermosyphons like those on the Trans-Alaska Pipeline [14] keep the ground frozen. |
| IMPACT | Zero-carbon heat and power for 45,000 people, no fuel deliveries, helium for the hospital's MRI scanners, and a converter that turns about 20 t of rubbish a day into gas and paving. | All tap water and about half of the protein made locally, about 20 t of $CO_2$ taken from the air every day, fewer cargo flights, and a 30-day food reserve. | Rest and health through the polar night, tourism and jobs, cafés serving food from the plant, and shelter for up to 1,000 people in an emergency. |

THE BIG IDEA

# Three Districts, One Loop

*The three signature districts do not work alone. Heat, water, oxygen and waste go round from one to the next instead of being thrown away.*




**Heat**

Spare heat from the fusion station melts the snow at the food plant, so the snow does the job of a cooling tower.

**Water**

The food plant's electrolysers leave slightly deuterium-rich water, which goes back to the fusion station as fuel.

**Oxygen**

The same electrolysers supply oxygen to the plasma waste converter and the hospital.

**Waste**

The converter turns rubbish into hydrogen and $CO_2$ for the food lines, and its slag becomes paving.

**Wellbeing**

The spa turns geothermal heat into rest and health, and its restaurants serve food from the plant.

EVERYDAY LIFE

# A Livable City Every Day



*Snowpine Residences at dusk. Sidewalks warmed by district-heating return water stay snow-free, and spruces shelter the courtyards from the wind.*

## HOMES

Family houses in Snowpine Residences, the western eco-homes, towers such as the Aurora 81 Residences and Ice Crystal Apartments, co-living, student and senior housing.

## LEARNING

Three kindergartens, Snow Lantern Primary School, the Northern Lights School for ages 6 to 18, and Arctic University.

## HEALTH

Snowflake General Hospital with about 250 beds and a 24-hour emergency department, a campus clinic and a 24-hour pharmacy.

## SHOPPING

Igloo Fresh Supermarket, Tundra Grocery, the night and fish markets and the Aurora Glacier Mall with its indoor atrium park.

## CULTURE AND PARKS

Aurora Cinema, a music hall, a library open 24 hours in the polar night, the Snow Stadium and parks with heated benches.



*The Mount Aurora Approach: low buildings keep the summit in view, and warm, low-glare lanterns do not hide the aurora.*

**250**
hospital beds

**900**
pupils at Northern Lights School

**5,000**
students at Arctic University

**8,000**
seats in the Snow Stadium



*The western eco-homes at dusk, with the Ice-Proof Wind Farm and the air base behind.*

HOUSING

# Homes for the Polar Night

*From passive eco-homes to towers with sky gardens, every home is built to stay warm, bright and safe at −45 °C, without thawing the ground beneath it.*

## HOW A HOME BEATS THE COLD

### Insulated piles

Homes stand on insulated piles above a ventilated air gap, so their warmth never thaws the permafrost.

### Passive-house shell

Thick walls, triple glazing and an airtight shell: an eco-home needs about 80% less heat than an ordinary house.

### Heat-recovery air

Fresh air is warmed by the stale air going out, saving about 80% of the heat that would escape.

### Winter-garden balconies

Glass-enclosed balconies stay usable at −45 °C and act as a warm buffer for the wall behind.

### Snow-shedding roofs

Steep, smooth roofs shed heavy snow by themselves, so they never overload or need shovelling.

### Light in the dark

Light tubes bring daylight deep inside, and dawn-simulating lamps wake families in the polar night.

## A HOME FOR EVERYONE

**600**
apartments in the Aurora 81 Residences

**350**
apartments in Ice Crystal Apartments

**150**
older residents in Golden Aurora Senior Residences

**120**
rooms in the Student Residence Hall

**12**
residents in the Snowpine Senior Care Home

*Plus family houses in Snowpine Residences and the Frostline Co-Living Tower for newcomers, young workers and visiting researchers.*

# Energy for the Polar Night



*Northern Renewable Park in the low sun: the Ice-Proof Wind Farm behind the Bifacial Solar Farm.*

*The sun disappears for two months, so the city uses sources that cover for each other.*

**Steady**

**Fusion**

Steady power and heat all year, even in total darkness.

**60 MW**

**Ice-proof wind farm**

Heated blades keep turning in freezing fog; most power comes in stormy winter.

**+25%**

**Bifacial solar farm**

Snow reflects light onto the panel backs. We plan for +25%; tests in snowy Canada measured about +19% [17].

**24/7**

**Geothermal plant**

Clean electricity and hot water around the clock, and the second heat source.

**200 MWh**

**Battery bank**

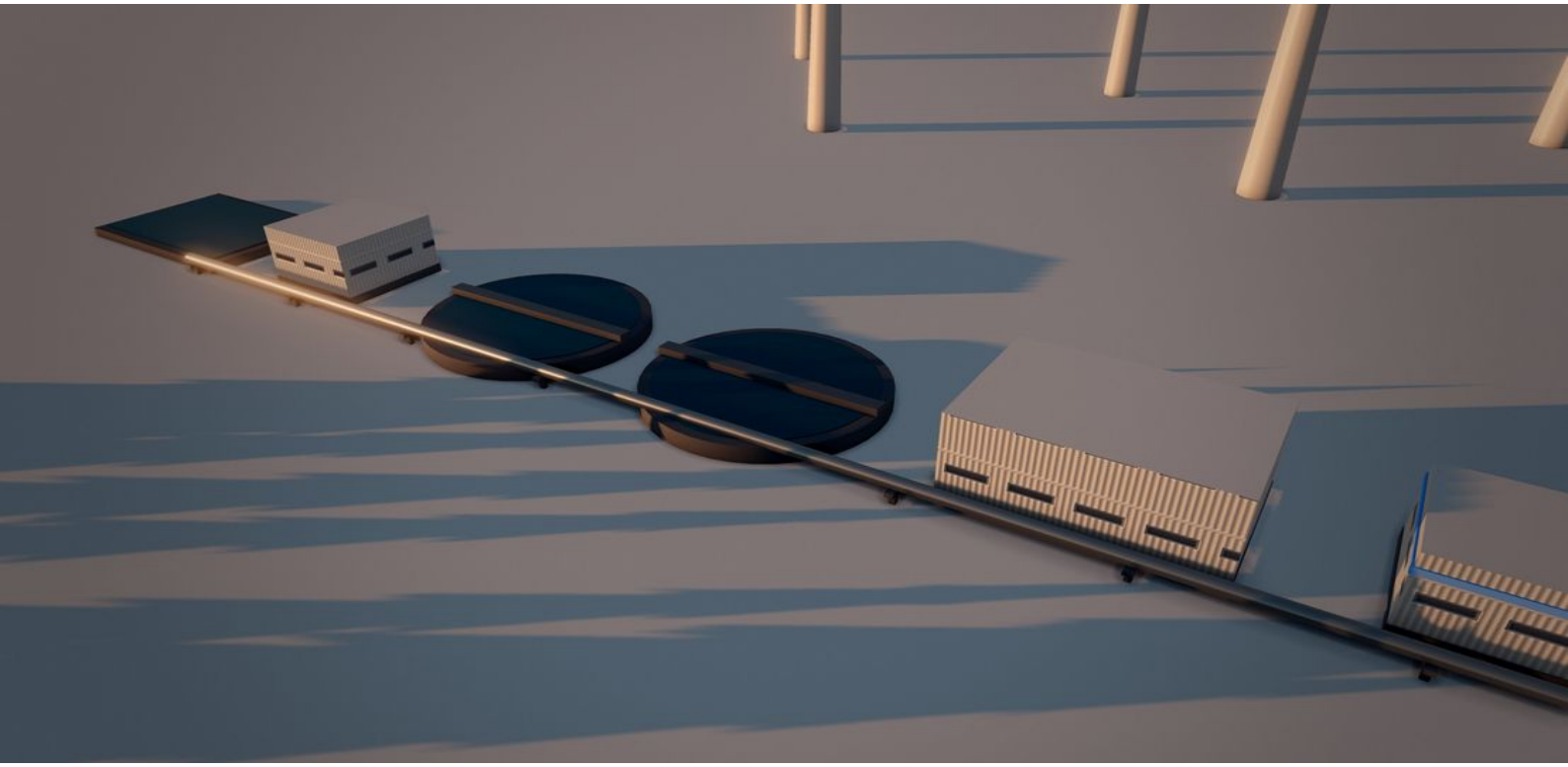
Recycled car batteries store spare wind and solar power for calm, dark days.

**12 h**

**Heat storage tank**

Keeps homes warm if a heat source stops, losing less than 1% of its heat a day.

# Nothing Is Wasted



*The six-stage Water Recycling Line in the Northern Renewable Park, with the wind farm behind.*

## THE SIX-STAGE WATER RECYCLING LINE

1. **Snowmelt & greywater intake**
2. **Intake pumps & bar screen**
3. **Settling basins**
4. **Sand & membrane filters**
5. **UV disinfection**
6. **Recycled water tower**

In Aurora Haven, water goes round and round. Clean snow becomes drinking water at the Snow-to-Food plant. Used greywater from homes flows to the Water Recycling Line: screens, settling basins, sand and membrane filters, UV light, and finally a tower that feeds toilets, laundries, greenhouses and fire hydrants.

Dirty street snow is melted separately and never reaches the taps.

The Circular Recycling Center sorts about 30 tonnes of materials a day with AI optical sorters and robot arms. Food scraps become biogas, and the Plasma Waste Converter handles what is left, turning it into clean gas and building blocks.

Nothing goes to landfill in the permafrost.

**95%**
of water recycled

**30 t**
of materials sorted a day

**20 t**
of rubbish a day turned into gas and blocks

**0**
landfill in the permafrost



*The eco-tour reindeer herd grazing on the tundra by the Snow Garden Park, with the foothill pines and Mount Aurora behind.*

URBAN ECO-SYSTEM

# Wild Neighbours

*Aurora Haven is built to share the tundra: animals can move through the city, plants work as snow fences, and farmed life closes the food loop.*

## ANIMALS

**Wildlife corridors.** Snow-laden trees on the eastern and western edges give snow buntings, ptarmigans, arctic foxes and hares cover between the open tundra and the parks. Roads beside them have dimmed lights and wildlife underpasses.

**Fences with gaps.** Park fences have open rails, so drifting snow blows through, and gaps at ground level, so foxes and hares pass freely.

**Reindeer.** The eco-tour herd is cared for by herders who follow Sámi traditions. Rides last only a few hours a day, and GPS collars help find animals grazing on the tundra.

**Polar bears.** Fibre-optic sensors in the air base fence alert a wildlife team when a bear comes close, so people and bears stay apart safely.

## PLANTS AND FARMED LIFE

**Living snow fences.** Belts of snow spruces and arctic pines trap drifting snow before it reaches the streets; the clean snow is collected for the food plant.

**Green stepping stones.** Snow lindens line Liberty Mall, and dwarf pines in heated planters on the Aurora 81 Sky Garden give birds a stepping stone across the city.

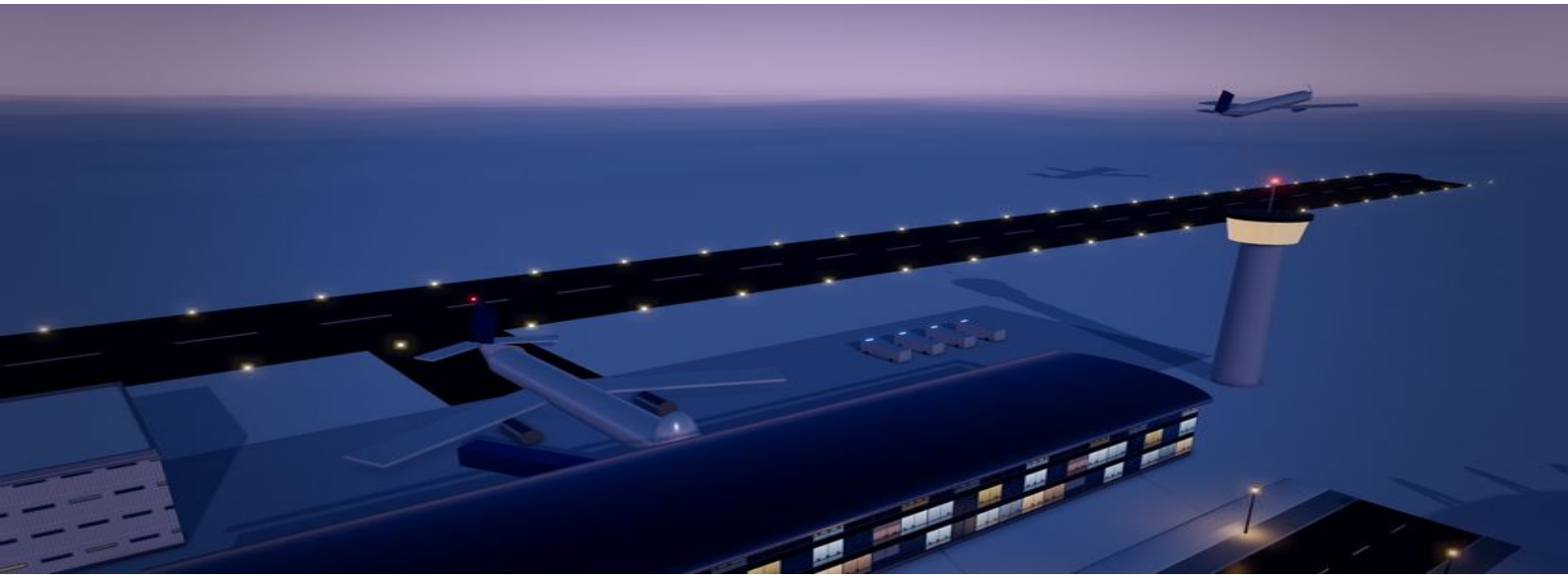
**Aquaponics.** In the Evergreen Aquaponics Dome, Arctic char and rainbow trout feed beds of lettuce, herbs and strawberries; the loop loses under 2% of its water a day.

**Shelter and science.** Rock gardens and boulders shelter ptarmigans and hares; nest boxes and a camera let schoolchildren count visitors with the Polar Climate & Ice Research Institute.

**HOW THEY SURVIVE** Arctic foxes grow a deep, pale winter coat that blends into the snow and keeps them warm down to about −50 °C [15]. The city helps them with cover, safe crossings and dark nights: outdoor lights switch on only when something moves.

TRANSPORT

# Getting Around Fossil-Free

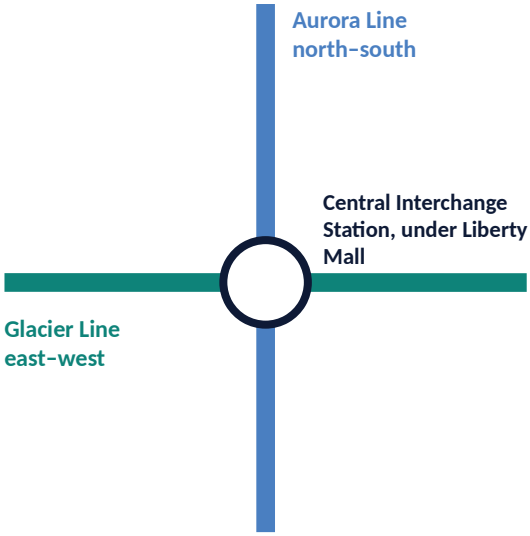


*Compact Polar Airport at dusk: the glass terminal, the control tower and Flight VN-81 climbing out.*

Transport in Aurora Haven is electric and built for the cold. The heart of it is the metro: the north–south Aurora Line and the deep east–west Glacier Line meet at the Central Interchange Station under Liberty Mall. The driverless four-car trains run underground, so blizzards never stop them.

On the streets there are electric trams, buses and shared robotaxis, and people wait in heated bus shelters. The roads are warmed from below, so no salt is needed, as in Reykjavík, where water returning from home heating melts the snow under streets [18].

The airport, with its heated runway and e-fuel planes, keeps the city supplied through deep winter.




### Users

Commuters, students, shoppers and stadium fans. Heated cars have space for wheelchairs, prams and skis.

### Capacity

Our estimate: if a four-car train carries about 400 people, a train every 3 min moves about 8,000 people an hour each way. After stadium events, trains every 90 s could clear the 8,000 seats in about 30 minutes.

### Safety

Deep, insulated tunnels kept at about +5 °C, emergency walkways with glow strips, and doors straight into the four shelters.

### Energy

All-electric and driverless. Braking returns about 30% of the energy, and insulation, not heaters, keeps the tunnels above freezing.

**3 min**
Aurora Line trains at rush hour

**90 s**
Glacier Line after big stadium events

**30%**
of train energy returned by braking

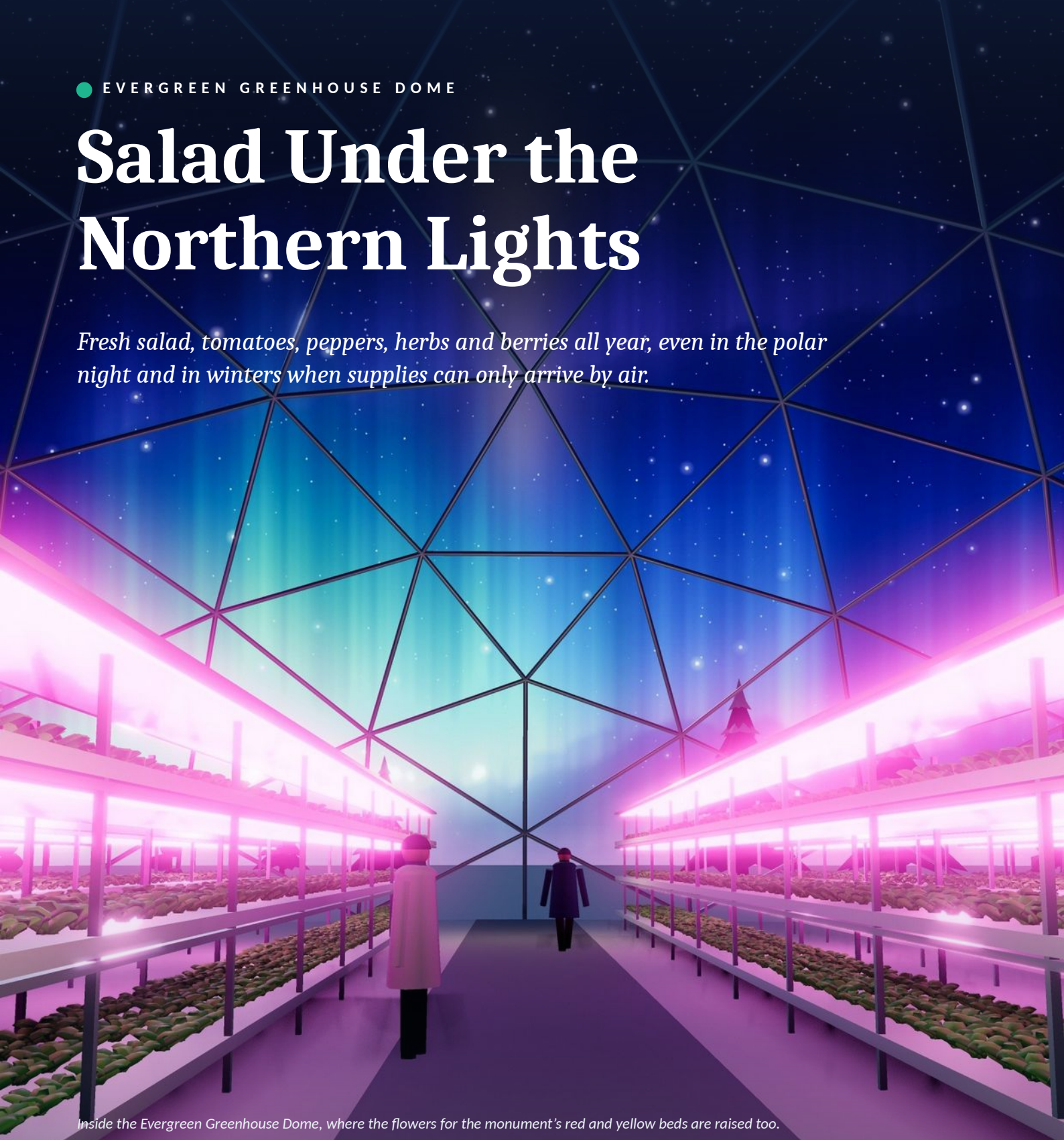
**2,800 m**
heated airport runway

**20 km/h**
downtown speed limit

EVERGREEN GREENHOUSE DOME

# Salad Under the Northern Lights

*Fresh salad, tomatoes, peppers, herbs and berries all year, even in the polar night and in winters when supplies can only arrive by air.*



*Inside the Evergreen Greenhouse Dome, where the flowers for the monument's red and yellow beds are raised too.*

The dome is heated by the district-heating loop and wrapped in thermal curtains at night. LED grow lights give the plants 18-hour "days" on stacked hydroponic racks, and $CO_2$ from the biogas plant and the waste converter boosts growth.

**18 h**

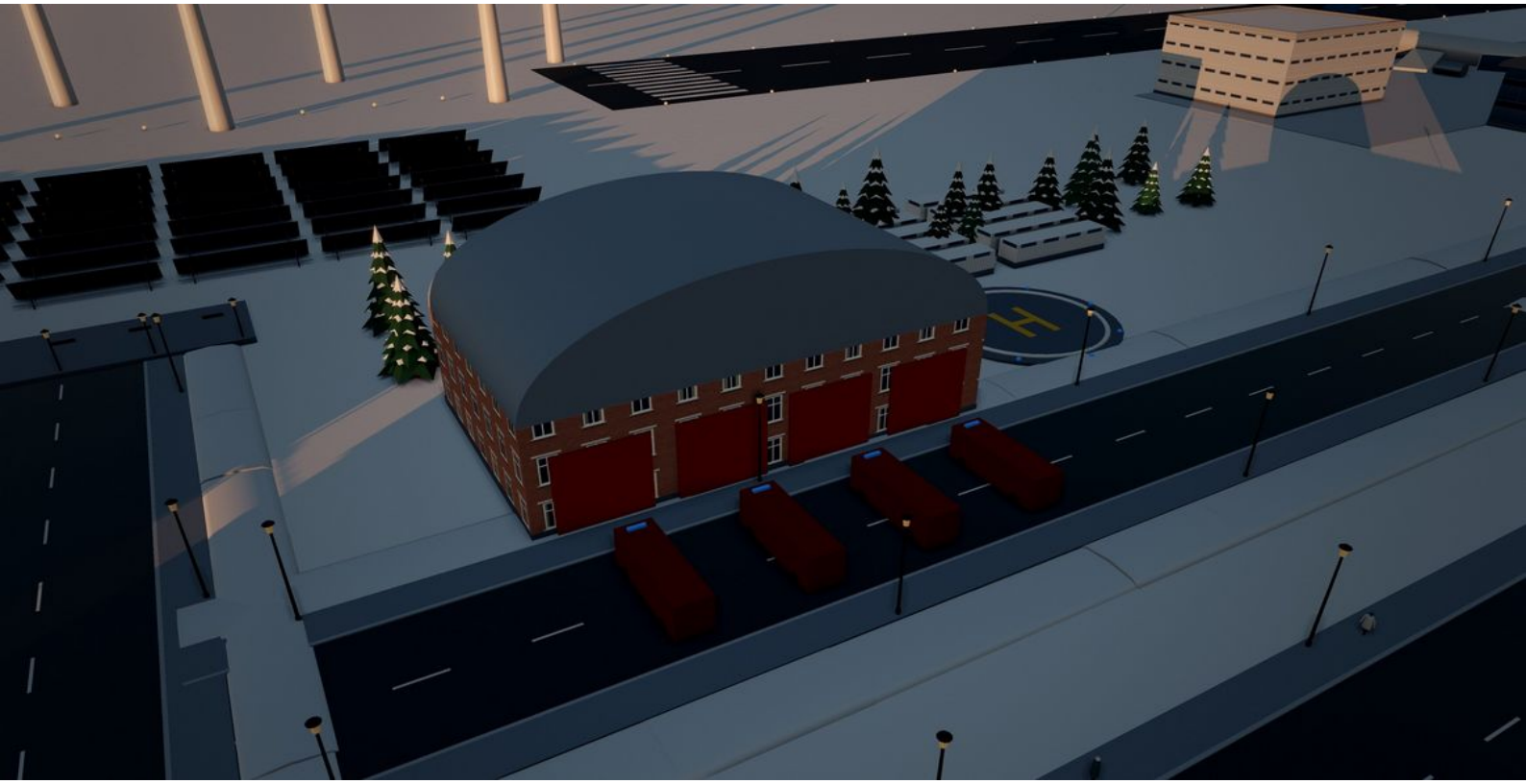
LED "days" for the plants

**90%**

less water than fields [16]

SAFETY

# Safe and Ready for Any Storm



*The Polar Rescue and Resilience Station in the Energy Hub, north of downtown, with its helipad.*

In the polar night, safety comes first. The Polar Rescue and Resilience Station is home to the fire brigade, ambulances, ice rescue and the 112 emergency centre, and its vehicles leave within 60 seconds.

The West Market Fire & Ambulance Station reaches any downtown address in about four minutes, and a rescue helicopter is ready day and night.

Under Liberty Mall, four underground shelters can keep every resident safe for 30 days. Aurora Core, the AI control tower, forecasts blizzards up to 48 hours ahead and alerts every phone.

Cafés and lobbies act as warm points, and the air base runway is a backup for supply flights.

**60 s**
for rescue vehicles to leave

**4 min**
to any downtown address

**10 min**
for the rescue helicopter to be airborne

**4**
underground shelters

**30 days**
of supplies for every resident

**48 h**
blizzard forecasts

# Pros & Cons

*No city is perfect. Here is what Aurora Haven does well, where it could fail, what it costs and what we gave up to make it work.*

## ADVANTAGES

**Zero carbon.** No fossil fuel burns anywhere; four clean energy sources cover for each other.

**Water and waste.** About 95% of water is recycled, and nothing goes to landfill in the permafrost.

**Food security.** All tap water and about half of the protein are made in the city.

**Safety.** Shelters for every resident for 30 days, 48-hour blizzard warnings and a metro that storms cannot stop.

**Wellbeing.** Daily needs within a 10-minute walk, plus a spa, a 24-hour library and parks for the polar night.

## LIMITATIONS AND RISKS

**Fusion is not ready.** ITER plans deuterium-tritium operation only in 2039 [4]; until fusion works, the city depends on geothermal, wind, sun and storage.

**One heating loop.** If it fails at −45 °C, homes cool fast; the 12-hour heat tank and the geothermal plant are the safety net.

**Thawing ground.** Any heat leak could crack foundations, as thaw already threatens Arctic infrastructure [2]; sensors must watch it constantly.

**Cut off by storms.** In deep winter all supplies come by air.

**High-tech risks.** AI control systems need cyber-security and skilled staff, and new foods need people's trust.

## COSTS

Building: a fusion plant is one of the most expensive machines ever made: ITER alone is estimated at €18–22 billion to first plasma [12]. Solar Foods' first protein factory cost about €42 million for up to 160 t a year [6]. Heated streets, insulated tunnels and triple glazing all cost more than ordinary ones.

Running: direct air capture uses a lot of energy and still costs hundreds of dollars per tonne of $CO_2$ [13]. Hydroponic lettuce needs about 82 times more energy than field lettuce, but about 92% less water and 11 times less land [16].

## TRADE-OFFS: WHAT WE CHOSE, AND WHAT WE GAVE UP

| | | |
|---|---|---|
| **Compact blocks that save heat** | → | smaller homes and less private space |
| **Metro, trams and shared robotaxis** | → | fewer private cars and less freedom to drive |
| **Lights that switch on only when needed** | → | darker streets, for the aurora and the wildlife |
| **Limited reindeer rides, protected corridors** | → | less tourist income, healthier animals |
| **Factory-made food** | → | no farmland needed, but more electricity and new tastes |

**OUR PLAN B** Build in stages: first the geothermal plant, the wind and solar farms, the battery bank and the heat tank; add the fusion station when the technology is proven. Until then, the city grows more slowly.



INVESTMENT OPPORTUNITY

# Invest in Aurora Haven

*The Mount Aurora Approach at dusk: the street from downtown to the spa.*

*Aurora Haven needs partners to build it. We are looking for investors who want a share in a zero-carbon city, built step by step, starting with technology that already works today.*

## THREE WAYS TO INVEST

### 1 Green city bonds

Fund Stage 1: the geothermal plant, the wind and solar farms, the battery bank and the heat tank, paid back from selling heat and power to the city.

### 2 Community shares

Residents and visitors buy shares in the wind and solar farms through the Aurora Green Finance Hall and share in the income they make.

### 3 Research partnerships

Companies and universities test new food, energy and building ideas with Arctic University and the Polar Climate & Ice Research Institute.

## WHAT YOUR MONEY BUILDS, STAGE BY STAGE

1. **Stage 1 · A clean base.** Technology that works today: the geothermal plant, the wind and solar farms, the battery bank, the heat tank, heated streets and the metro.
2. **Stage 2 · Food and water.** The Snow-to-Food & Water Plant, starting small. For scale, Solar Foods' first protein factory in Finland cost about €42 million for up to 160 t a year [6].
3. **Stage 3 · Fusion.** Built only when fusion power is proven. It is the most expensive part: ITER alone is estimated at €18–22 billion to first plasma [12], which is why it comes last.

## WHAT INVESTORS GET BACK

- Steady income from heat, power, water and food sold to 45,000 residents
- Visitors to the Mount Aurora spa and the Snow Stadium all year
- Know-how other Arctic towns can use: about 4 million people live in the Arctic [3]
- A city with no fuel to import and no carbon to pay for

*Please note: these are future concepts, not tested projects. Costs and returns still have to be worked out in a full business plan, and no return is guaranteed. The risks are on page 33.*

**READY TO BUILD THE FUTURE?** Visit the Aurora Green Finance Hall in the Financial District, or meet our team on page 5.

EXPLORE

# Take the Tour: Eight Stops

*A walk around Aurora Haven in eight stops, from the civic heart of downtown out to the plants, the snow peak and the windy northern edge.*

1. **Founding Mayor Monument**
   The civic heart of the city on the Aurora Roundabout.
2. **Metro Entrance – Central Interchange Station**
   Escalators down to the metro and the shelters.
3. **Snowflake General Hospital**
   About 250 beds and a 24-hour emergency department.
4. **Fusion Core – Compact D-T Tokamak**
   The city's own sun.
5. **Snow Harvest & Melt Hall**
   Melts about 3,000 t of snow a day with spare fusion heat.
6. **Mount Aurora – Snow Peak & Hanging Glacier**
   The 35 m peak with the spa inside.
7. **Ice-Proof Wind Farm**
   Two rows of turbines making about 60 MW.
8. **Polar Rescue and Resilience Station**
   Home of the 112 emergency centre.

*If there is time: the Aurora Lagoon, the Bifacial Solar Farm and the Recycled Water Tower.*



*Aurora Snow Stadium at dusk.*

BEHIND THE BUILD

**35**
key places, numbered on the master plan

**9**
downtown blocks around the Aurora Roundabout

**1 m**
per unit on every plan, drawn from the 3D model

# Places to Visit

*The numbers match the markers on the city master plan on page 12.*

**1 Founding Mayor Monument**
The city's civic heart on Aurora Roundabout: ceremonies, national holidays and walking tours.

**2 Aurora Civic Center & Municipal Hall**
Town hall for the city council, ID cards, permits, housing and family services.

**3 Snowflake General Hospital**
Main hospital with about 250 beds and a 24-hour emergency department.

**4 Northern Lights School & Botanical Conservatory**
Primary and secondary school for about 900 pupils aged 6–18.

**5 Igloo Fresh Supermarket & Food Hall**
Downtown's main supermarket and food court; about half its food is made in the city.

**6 Aurora Cinema & Performing Arts Center**
Cinema and theatre for films, concerts, ballet and school shows.

**7 Underground Metro & Emergency Shelters**
Metro interchange linked to four shelters that last 30 days.

**8 Aurora Core – AI Control Tower**
Operations centre where an AI digital twin runs traffic, heat, power and alerts.

**9 Aurora Haven Police Headquarters**
Central police station with a 24-hour police control room.

**10 Arctic University – Main Hall**
Lecture halls for about 5,000 students of polar engineering, fusion, food science, medicine and climate.

**11 Aurora Public Library**
Free library, open 24 hours in the polar night; lends skis, telescopes and light-therapy lamps.

**12 West Market Fire & Ambulance Station**
Downtown fire and ambulance branch that reaches any downtown address in about 4 minutes.

**13 Aurora Glacier Mall & Crystal Atrium**
Largest downtown mall, with an indoor atrium park of real trees.

**14 Snow Lantern Primary School**
Primary school for about 450 children aged 6–11 from the southern districts.

**15 Pinecone Kindergarten**
Kindergarten and day nursery for about 40 children aged 1–6.

**16 Aurora 81 – Polar Landmark Tower**
Tallest tower and skyline symbol: offices, hotel and sky apartments, net-zero energy.

**17 Aurora Snow Garden Park**
Public park with heated benches, a sledging slope and a snow-sculpture trail.

**18 Mount Aurora – Snow Peak & Hanging Glacier**
35 m landmark peak with the four-level spa inside.

**19 Summit Sanctuary & Aurora Eye**
Float pool, meditation circle and the Aurora Eye viewing bubble for the northern lights.

**20 Aurora Lagoon – Outdoor Geothermal Mineral Pool**
38 °C open-air silica pool for bathing under the northern lights.

**21 Fusion Core – Compact D-T Tokamak**
150-million-degree plasma giving about 800 MW of heat; it cannot melt down.

**22 Power Conversion & Heat Export Hall**
Makes about 300 MW of electricity and sends about 450 MW of heat to the city.

**23 Plasma Waste Converter**
Turns about 20 t of rubbish a day into clean gas, recovered metals and building blocks.

**24 Snow Harvest & Melt Hall**
Melts about 3,000 t of clean snow a day with spare fusion heat.

**25 Drinking-Water & Mineral Works**
Makes about 2,500 $m^3$ of drinking water a day, all of the city's tap water.

**26 Bioreactor Silos**
Six silos where microbes make about 2.5 t of protein flour a day.

**27 Ice-Proof Wind Farm**
Two rows of ice-proof turbines making about 60 MW.

**28 Bifacial Solar Farm**
Double-sided panels that make power 24 hours a day in the midnight-sun months.

**29 Water Recycling Line – Recycled Water Tower**
Stores recycled water for toilets, laundries, street cleaning, irrigation and hydrants.

**30 Geothermal Power Plant**
Clean electricity and hot water around the clock; the second heat source.

**31 Polar Rescue and Resilience Station**
Fire, ambulance and ice-rescue headquarters, home of the 112 emergency centre.

**32 Glass Terminal, Control Tower & Gate 1 Dreamliner**
Passenger and cargo airport; the city's supply link in deep winter.

**33 Evergreen Greenhouse Dome**
Grows fresh vegetables and berries all year under LED light.

**34 Circular Recycling Center**
Sorts, repairs and recycles about 30 t of materials a day.

**35 Aurora Snow Stadium**
About 8,000 seats for snow sports, school sports days and the Midwinter Aurora Festival.

KEY FIGURES

# Aurora Haven by the Numbers

| | | | |
|---|---|---|---|
| **THE CITY** | **45,000**<br>residents | **9**<br>downtown blocks | **35**<br>key places |
| **CLIMATE** | **−45 °C**<br>winter lows | **2 months**<br>of polar night | **120 km/h**<br>blizzards |
| **FUSION POWER STATION** | **300 MW**<br>electricity | **450 MW**<br>district heat | **99.8%**<br>heavy-water purity |
| **SNOW-TO-FOOD & WATER PLANT** | **5 t**<br>hydrogen a day | **4 t**<br>starch a day | **1 t**<br>fat a day |
| **ENERGY AND STORAGE** | **60 MW**<br>wind farm | **200 MWh**<br>battery bank | **<1%**<br>heat lost a day by the heat tank |
| **MOUNT AURORA SPA** | **4**<br>levels, 0 to 13.5 m | **28 °C**<br>all winter | **1,000**<br>people sheltered in an emergency |
| **LIVABILITY** | **600**<br>Aurora 81 apartments | **450**<br>pupils at Snow Lantern Primary | **3**<br>kindergartens |
| **TRANSPORT** | **2**<br>metro lines | **5 min**<br>spa shuttle | **20 min**<br>airport shuttle |

*All figures are design values for the Aurora Haven city project; page 38 explains how to read our numbers.*

# Sources & Assumptions

## HOW TO READ OUR NUMBERS

### Design values

Figures for our own city plan, such as 300 MW of electricity or 3,000 t of snow a day. They come from our design, not from a real city.

### Our estimates

Calculations we made ourselves and marked "our estimate", such as the metro capacity, or worked out on page 23. They show an idea is reasonable, not exact.

### Future concepts

The fusion station and the snow-to-food plant do not exist at this size today. They are our team's ideas for the future; no prototype has been tested.

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**PICTURES** Every picture, plan and diagram in this magazine was made from our own 3D model of Aurora Haven (built with three.js). Text facts about the city come from our city plan and its place cards.

**AI USE** We used AI tools to help us build the 3D model of the city and make the pictures, plans and diagrams in this magazine, and to check our spelling and fix logic errors in the text.

JOIN US

# Warm, Green and Ready

As we conclude this feature on Aurora Haven, we are left with more than just an impression of futuristic design – we are reminded of our shared responsibility toward the planet. The project clearly demonstrates that advanced technology and nature can coexist in harmony when guided by vision and innovation. Aurora Haven not only sets a precedent for resilient urban planning in extreme climates, but also serves as a powerful inspiration for a sustainable future where human communities are built to endure. May the concepts presented here spark further dialog and inspire bold solutions for generations to come.

**Be one of the first polar citizens.** Buy a community share in our wind and solar farms through the Aurora Green Finance Hall, study polar science at Arctic University, or come for a night at the Mount Aurora spa.

## AURORA HAVEN

*A polar city that thrives*

### CONTACT THE TEAM

**Polar City**
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SCAN TO READ

**The interactive edition**

Flip through this magazine online on FlipHTML5
Open the flipbook

